

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Weather Travel Advisory Agent | API-Workflow Domain (Events & Travel Desk)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external weather forecast API integration

1. Problem Statement

Event coordinators and traveling employees frequently email each other asking whether the weather will be suitable for an upcoming outdoor event or business trip — for example, checking if rain is expected on the day of a campus outdoor event, or whether a destination city will be too hot or stormy for a planned site visit. Today, staff manually check a weather app or website and reply by email — a small but repetitive task that delays planning decisions, especially when several events or trips are being coordinated at once.

Design and build an Autonomous UiPath Agent, the AI Weather Travel Advisory Agent, that reads an incoming email asking about the weather for a given city and date, calls a live Weather Forecast API workflow to fetch the forecast, assesses suitability for the stated purpose, and independently replies with an advisory — without any human approval step.

2. System Prompt

You are an AI Weather Travel Advisory Agent for the Events and Travel desk. Your goal is to analyze weather advisory request emails and provide an accurate, up-to-date forecast and suitability assessment using a live weather forecast API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Event Weather Advisory (outdoor event on a specific date)
 - Travel Weather Advisory (trip to a city on a specific date)
 - Other
3. If it is Event Weather Advisory or Travel Weather Advisory:
 - Extract the city and the date (or date range)
 - Call the Weather Forecast API workflow with the city and date
 - Assess suitability: flag as 'Caution' if heavy rain, storms, or extreme temperature is forecast; otherwise 'Favorable'
 - Generate a professional reply stating the forecast summary, temperature range, precipitation chance, and the suitability assessment
4. If it is Other:
 - Skip processing

```
Rules:
- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the city or date is unclear, ask
  the requester to clarify instead of guessing
- Do not invent forecast values – always use the API response
- Return output in structured format
```

3. User Prompt

```
Analyze the following weather advisory request email and provide the
forecast and suitability assessment using the live weather forecast API:
{{weather_email_text}}
```

4. Input Schema

```
{
  "weather_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "city": "string",
  "forecast_date": "string",
  "forecast_summary": "string",
  "temperature_range": "string",
  "precipitation_chance": "string",
  "suitability": "Favorable | Caution",
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- city (text)
- forecast_date (text) — format yyyy-MM-dd

Configure the workflow to call a live weather forecast API endpoint (e.g., an HTTPS GET request to a weather-data provider) and parse the JSON response for forecast summary, temperature range, and precipitation probability.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
city	location_query
forecast_date	forecast_date

8. Updated System Prompt — Tool Invocation Logic

If the email is classified as Event Weather Advisory or Travel Weather Advisory:

- Call the `get_weather_forecast` API workflow with `city` and `forecast_date`.
- Parse the API response for `'summary'`, `'temp_min'`, `'temp_max'`, and `'precipitation_chance'`.
- If `precipitation_chance` is high or an extreme temperature is forecast, set `suitability` = "Caution"; otherwise set `suitability` = "Favorable".
- Use the `generate_reply` GenAI activity to draft a professional reply email with the forecast summary, temperature range, precipitation chance, and suitability assessment – fully autonomously, with no manual approval required.

If the email is classified as Other:

- Take no further action.

9. Tools & Activities Summary

Tool / Activity	Type
<code>get_weather_forecast</code>	API Workflow (HTTP Request to a live weather forecast API)
<code>generate_reply</code>	GenAI Activity (LLM prompt-based text generation)
<code>classify_request</code>	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live weather forecast API, assesses suitability, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external weather API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and derive a suitability assessment from the returned forecast
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder